


The Effects of the New Jersey-Pennsylvania Negative Income Tax Experiment on Health and Health Care Utilization
Author(s): David Elesh and M. Jack Lefcowitz
Source: *Journal of Health and Social Behavior*, Vol. 18, No. 4 (Dec., 1977), pp. 391-405
Published by: American Sociological Association
Stable URL: http://www.jstor.org/stable/2955347
Accessed: 12-12-2015 00:38 UTC

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at http://www.jstor.org/page/info/about/policies/terms.jsp

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.



Sage Publications, Inc. and American Sociological Association are collaborating with JSTOR to digitize, preserve and extend access to *Journal of Health and Social Behavior*.

http://www.jstor.org

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

THE EFFECTS OF THE NEW JERSEY-PENNSYLVANIA NEGATIVE INCOME TAX EXPERIMENT ON HEALTH AND HEALTH CARE UTILIZATION *

DAVID ELESH
Temple University

M. JACK LEFCOWITZ
The University of Wisconsin–Madison

Journal of Health and Social Behavior 1977, Vol. 18 (December):391–405

This paper examines the effects of the recently completed New Jersey-Pennsylvania Negative Income Tax Experiment on health and health care utilization. Both economic and sociological theory suggest that the negative tax benefits may, in the short run, increase the proportions reporting themselves in ill health and utilizing care, and, in the long run, reduce these figures. However, the empirical literature suggests the effects may be modest, vary by family members, and be swamped by other variables. Data are presented for two or three points in time (depending on the family member) over the three year duration of the experiments and no effects of any kind are observed. The results give credence to suggestions that health and health care utilization may be more a function of life style or preferences than income. While neither economic nor sociological theory asserts the effects of income are simple, these data suggest its theoretical role is still not well understood.

This paper examines the effects of the recently completed New Jersey-Pennsylvania Negative Income Tax Experiment on health and health care utilization.[1] The issue is an important one for two reasons. First, for those family members who are potential or active labor-force participants, the experimental benefits, by lowering the costs of not working and increasing the funds available for health care, may lead individuals to acknowledge health conditions that their need to earn had previously caused them to ignore. Second, the experimental payments may make possible expenditures for health which improve the ability to work or to engage in other, nonwork activities. Thus, the experiment's effects on health and the utilization of health care may have an important bearing on the assessment of the costs and benefits of a negative income tax program.

The conditions under which these possibilities arise can be phrased in terms of both economic and sociological theory. Both types of theory begin by drawing a distinction between the theory appropriate to predictions regarding experimental effects on health itself and the theory appropriate to predictions regarding the effects of the experiment on the utilization of health care. Economic theory stresses the effects of income, while sociological theory, which generally accepts the economic model, elaborates the role of a number of factors recognized but little investigated by economists.

ECONOMIC AND SOCIOLOGICAL THEORY

Economic Theory of Experimental Effects on Health

Predictions as to the effects of the experimental payments on health follow from a simple model of the allocation of time based on the general theory of choice. The unit of analysis is defined as the household, made up of individuals whose preference functions are combined to form the aggregate household preferences and whose activities determine its resource base. For those household members who are actual or potential suppliers

* Names of authors are listed alphabetically. This paper has benefited from the comments of the *Journal's* referees; they are, of course, not responsible for any errors that may remain. The research reported here was supported in part by funds granted to the Institution for Research on Poverty at the University of Wisconsin by the Office of Economic Opportunity pursuant to the Economic Opportunity Act of 1964.

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

of labor, the fundamental time allocation problem lies in the decision as to the amount of time spent working by each family member. The factors affecting this decision are fourfold: (1) the family's views or preferences; (2) the expected market earning rates of its members; (3) the nonmarket earning rates of its members; and (4) the household's total budget constraint. This last factor is a function of the total number of hours available, the set of prospective market and nonmarket earning rates, any other income the family expects to receive, and the monetary value of its saleable assets. The experimental payments could affect the health of the labor suppliers through all four of these factors.

If the family treats the benefits as additions to what Friedman (1957) has termed their "permanent" or long-run average income—the income on which they base their allocation decisions—then, in the short run, the payments increase the nonmarketing earning rates of the family members relative to their market rates and increase the household's total budget constraint. By thus reducing the costs of not working, the payments may lead some family members not previously willing to acknowledge a health condition to do so, thereby increasing the proportion of persons reporting themselves as in ill health. In addition, other family members, who dislike work and who now find the cost of not working reduced, may become "ill" in order to have a legitimate reason for diminishing their labor supply (Cole and LeJeune, 1972). More generally, the payments may affect the health of all family members (including those who do not supply labor) by increasing the funds available for the utilization of health care, perhaps promoting some individuals (previously inhibited from doing so by their financial inability to afford care) to acknowledge health conditions.

In the long run, however, the proportion of those who were encouraged to declare themselves ill by their greater ability to afford care and the proportion who had otherwise declared themselves ill should decrease in relation to the efficacy of the care received.[2]

Economic Theory of Experimental Effects on Utilization of Care

As is implicit in the above discussion, the effects of the experimental payments on the utilization of health care follow from the theory of consumption. Within this theory, consumption, other things being equal, is considered a function of the household's preferences which are, themselves, typically taken to be a function of the household's permanent income. In other words, at different levels of permanent income, families become accustomed to different expenditure levels for different kinds of goods and services and their preferences change accordingly. The implication of these propositions for utilization of care is that if health care is a normal good—that is, if consumption of it increases with income—then the additional income represented by the payments should increase its utilization—this increased utilization being, in the short run, primarily attributable to the accumulation of health problems that the family could not previously afford to have treated. In the long run, however, the experiment's effects on utilization depend upon whether the families perceive the payments as additions to permanent or to transitory income. If they see the benefits as additions to permanent income, then they can be expected to adopt the higher levels of consumption typical of these higher income levels. On the other hand, if they see the payments as additions to transitory income, then utilization should decline as the backlog of untreated health conditions is reduced.

Sociological Theory of Experimental Effects on Health and Health Care Utilization

Given the *ceteris paribus* assumptions made in the above economic theories, sociological theory would predict the same effects, but would be chiefly directed to analyzing effects when those assumptions are not made. Sociological analysis of the relationship between health and utilization of care begins with the

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

recognition that health is socially defined: that is, individuals and groups differ in their ability to recognize objective symptoms, in terms both of what they define as symptoms of illness and of the interpretations they place upon them. Depending upon their situation and personal characteristics, individuals may or may not be aware of their symptoms, may or may not define these symptoms as being indicative of illness, may or may not consider themselves sufficiently ill to require medical attention, and may or may not obtain that attention. And, again, depending on their situation and personal characteristics, they may be selective in where they seek that attention. Thus, the process may be seen as a series of decision points leading to use of the medical care system with a variety of factors affecting each successive step (Andersen, 1968, Mechanic, 1968).

Among the most important of the factors that have been shown to affect the elements of this process are the individual's values, ethnicity, education, income, age, sex, and family size (Aday and Eichorn, 1972, Andersen et al., 1975). Within the context of the array of these variables, the effects of income are necessarily limited by the extent to which income is correlated with or, more importantly, a function of the other variables. In particular, an individual's income is clearly a function of his age, sex, ethnicity, education, and, perhaps, values. Thus, the effects of income on health attitudes or behaviors can be expected to diminish commensurate with the extent to which these causally prior variables are held constant.

The available evidence suggests that, given such controls, income has little or no net effect on most health variables. For example, Feldman (1966) and Samora et al. (1962) found that when education is controlled, most of the differences in health knowledge due to income disappear; but when income is controlled, the differences by education remain. The findings with respect to illness are more difficult to assess because of inadequate controls in the available data and because of some ambiguity as to how much of the association between health measures and income is to be attributed to the lower earnings of the unhealthy rather than to an increase in feelings of health, better diet, better housing, or to the higher use of medical care attributable to income. However, Lefcowitz (1973a) has pointed out that with suitable controls for age, or age and sex, the differentials by income for a larger variety of illness conditions become highly attenuated or disappear altogether. Similarly, Luft (1972), in his review of the literature, finds a frequent absence of any income effect on illness; and where the literature does support an income effect—as is the case for injury rates, number of chronic conditions, number of restricted activity and work days lost—one must ask whether the observed effects are due to the occupational hazards faced by and greater physical demands placed upon low-income workers rather than to income itself. Unfortunately, there has been little research on this question. More generally, there has been little research on the relative effects of health on income and income on health, although recent assessments of the existing evidence suggests that the effects of health on income are by far the more important of the two (Harkey et al., 1976; Lefcowitz, 1973a; Luft, 1972; Andersen and Benham, 1970).

A more persuasive argument can be made for the effects of income on utilization because, whatever effects other variables may have on the propensity to seek care, its actual consumption would seem to depend on the family's ability to pay for it. Yet, here again, the effects of income are unclear and are contingent upon the measure of utilization one employs. The National Health Survey (National Center for Health Statistics, 1972a) has found differentials by income for the average length of hospital stays, the rate of hospital discharges, and the rate of surgically related discharges, but for persons above the age of sixteen has found no differences in the number of physician visits per person per year. (For children below the age of 17, income has a substantial effect; children in families with annual incomes under $3,000 average 2.7 physician visits

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

per person per year while children in families with incomes of at least $10,000 average 4.3.) We must note, however, that income effects disappear when head's education or education and age is taken into account (Benham and Benham, 1975; Lefcowitz, 1973b). Similarly, Shortell (1975), analyzing the total number of ambulatory care visits for major illness episodes, found income to have no measurable net effect after controlling for other patient, provider, and source of payment variables.

While it would seem from the above review that the effects of income are uncertain and probably small, it can be argued that the literature has inadequately examined them. First, a serious methodological problem arises from the fact that most of the predicted income effects are predicated on a family's permanent income, but the existing estimates are based on measures that include transitory income. Measured income—because of episodes of unemployment, illness, or on the other hand, temporary good luck—is typically different from permanent income, so the literature has (to some degree) misestimated the real underlying influence of income. The second argument—that studies frequently fail to control for the possession of health insurance—leads to the conclusion that the effects of income are underestimated only if it is presumed that insurance reduces the differentials to be expected by income. However, because income and possession of health insurance are highly correlated (National Center for Health Statistics, 1972b), this presumption may be unwarranted. For example, Richardson, in his study of the use of physicians by a low-income population, found that whether considered separately or together, family income and health insurance had no effect on their use of physicians' services (Richardson, 1971).[3]

In summary, expectations as to the effects of the experimental benefits on health and health care utilization are by no means clear. While the predictions of economic theory may hold if the families treat the payments as additions to their permanent income, the magnitude of these effects relative to those of education, age, ethnicity, values, and other factors may be so small as to be unobservable.

Apart from the general question as to whether the experiment is likely to affect health and the utilization of care, we may also ask whether effects are more probable for some family members than for others. With regard to medical care, Koos (1954) and Richardson (1971) have suggested that use should be related to the instrumentality of a family member's position in the household because the cost of illness should increase with instrumentality. Since the husband is typically the main earner, this would imply that experimental effects would be most probable for him. Wives would be the next most likely to be affected, since they are the next most instrumental, and the children would be least affected. One may make a similar argument with respect to the effects of the experiment on health of the family members: those who are most instrumental should be least likely to become unhealthy because the costs of doing so are greatest.

On the other hand, there is another line of argument which suggests that the experimental payments may have their primary effect on the use of care by children—particularly children under five or six years old. According to this view, children are most likely to benefit from the payments for two reasons. First, the high socioemotional investment their parents have in them ordinarily leads the parents to seek care at the first signs of illness, and the payments should reinforce that tendency. Second, children, especially young children, in addition to being confronted with many of the diseases of their elders, are subject to a variety of diseases such as measles, chicken pox, and mumps which generally involve the use of medical care; thus, their greater need for care may interact with the presence of the payments to produce greater use of care for children and may, under certain definitions of ill health, produce more ill health among them. For example, parents, who (given the payments) are better able to afford the costs of unhealthy children, may be more willing to accept certain symptoms as indicative of illness and more willing to seek care than they had been previously.

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

Since there are no a priori grounds on which to decide between these two lines of argument, we must turn to the literature for assistance. Unfortunately, there are almost no pertinent studies. Indeed, perhaps the only systematic research in this area is Richardson's (1971) study of the use of physicians by a low-income population, which showed that the relationship between family status and delay in seeking care or amount of care received was contingent on the seriousness of the illness episode. When an illness was not serious—which is to say, most of the time—children and particularly children under six received care more quickly and had more of it than either of their parents; however, when the illness was serious, the male head of the household was the quickest to receive it and consumed more of it. Wives occupied a somewhat intermediate position: regardless of the seriousness of their illness, they delayed the longest in seeking care; and the amount of care they received was about the same as that of their husbands when their illness was nonserious, and intermediate between the amounts received by their husbands and children when it was serious. Thus, from this evidence, it would appear that instrumentality determines the use of care when an illness is serious; otherwise, what appears to be a generalized parental concern for young children does.

Finally, we may ask whether, aside from the possible general effects of the payments on health and utilization or differential effects according to status in the family, there are effects on the pattern of care. For example, there is some literature to suggest that people prefer private physicians but use "public" physicians when they cannot afford the former (Elesh and Schollaert, 1972). Thus, it is possible that with or without an increase in utilization, the experimental payments may cause a shift toward greater use of private physicians.

THE EXPERIMENTAL DESIGN[4]

The primary purpose of the experiment was to assess the labor supply response of the urban poor in intact families, and consequently, the sample was restricted to families (1) with at least one dependent person and an able-bodied male who was between 18 and 58 years of age and not in the armed forces, a student, or institutionalized and (2) with total family income no more than 150 percent of the poverty line for each family size. The sample was drawn from poverty tracts in Trenton, Paterson, Passaic, and Jersey City, New Jersey; and Scranton, Pennsylvania.

Once eligibility was determined from a special screening interview, families were assigned to one of eight negative tax plans, which together defined the experimental group, or to the control group. The eight tax plans were combinations of guarantee levels and tax rates which, in our judgment, encompass the area of greatest policy interest. Guarantee levels are the annual values of the experimental benefits paid when family income is zero.[5] Tax rates are the rates at which the benefits are reduced as family income rises. Four guarantee levels (50, 75, 100, and 125 percent of the poverty line for each family size) and three tax rates (30, 50, and 70 percent) were tested. The eight tax plans involved the following combinations of guarantee levels and tax rates: 50–30, 50–50, 75–30, 75–50, 75–70, 100–50, 100–70, 125–50. Assignment of the experimental and control groups utilized an optimal design model developed by Conlisk and Watts (1969), which minimized, subject to certain budgetary constraints, the error variance of the estimates of changes in family earnings (adjusted for family size) induced by the experimental payments. To have attained comparable precise estimates with a purely random assignment would have required a substantially larger sample at much greater cost. However, use of the model meant that all of the experimental plans differed in average family income and required special analytic controls which are discussed below. Seven hundred twenty-five families were assigned to the experimental group and 632 to the control group, and the sample was stratified by the three major ethnic groups in the experimental sites—whites, blacks, and Spanish-speakers—and by three in-

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

come strata, where income was weighted by family size.

After the families were assigned, all (experimental and control) received a pre-enrollment interview to obtain baseline data in a variety of areas uncontaminated by knowledge of the experiment or the inception of the negative tax transfers. Subsequently, the three year program was explained to the experimental families and those who agreed to participate (less than seven percent refused) began receiving payments every two weeks. Their obligation was to report their income and family composition each month. Both the experimental and control groups were interviewed quarterly.

METHODOLOGICAL ISSUES

In an earlier section, we commented that the effects of the experiment on health and the utilization of health care may be contingent on the particular measures of these factors employed. In this section, we address this question in greater detail, discussing in turn the kinds and quality of the data on health and utilization of care, the timing of their measurement, the parameterization of the experimental plans, and the operationalization of the sociological model for the estimation of experimental effects. Following this, we describe the sample analyzed here.

Data

Because of the presumed importance of health on labor-supply response and as a response itself to the experiment, a substantial amount of health data were gathered for the head, spouse, and children of the household. The available data include, for the year prior to the interview, measures of the number and type of chronic illnesses, the number of days spent in a hospital, the number of hospital stays, the number of days spent in bed because of illness, the number of days of work (including housework for wives, school for children) lost due to illness, and the number of visits to physicians of various types. The range of these data is sufficiently broad for one to feel confident that any reasonable definition of health or ill health and the most important types of health care utilization will be captured within it, and the information is similar to that collected in many other health surveys (indeed, most of the questions on health were taken from the National Health Survey). It may be, however, of somewhat better quality, in that the head and spouse were each asked to report on his or her health and utilization of care (one parent reported on their children's health), whereas the more common practice is to ask a single informant to report on the health and health care utilization of the entire household.

From these data, we shall examine the effects of the experiment on the head's and spouse's number of chronic conditions, number of work days lost, number of days spent in a hospital, and number of physician visits both in total and decomposed into private and "other" visits; for the children, we shall examine the experiment's effects on the per capita number of chronic conditions, the per capita number of days spent in bed, whether any child has spent at least one night in a hospital in the year previous to the interview, and the per capita number of visits to a physician similarly categorized. For present purposes, we shall ignore possible experimental effects on the adults' types of chronic conditions, number of days spent in bed, and number of hospital stays, and on the children's types of chronic conditions, number of days lost from school, and number of days spent in a hospital. The reason for the exclusion of types of chronic conditions is that our primary interest is in whether any chronic condition appears or disappears as a result of the experiment, not in specific types of conditions. Days in bed is excluded from the adults' analyses because it is highly correlated with days lost from work (average r at a given time point = .85),[6] and the latter is more likely to capture something of the family's loss in welfare due to lost income. The number of adult hospital stays is not analyzed simply because there is little variance in the variable and it is highly correlated with the number of days spent in a hospital (average r at a given time point = .94). The number of days lost

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

from school was dropped because it does not sufficiently indicate the ill health of the preschoolers and because, in any event, it is highly correlated with per capita days in bed. Finally, the reason that we did not analyze children's per capita hospitals days is that, because so few children are hospitalized, the variable is extremely skewed. Consequently, a simple dichotomous measure of hospitalization was employed. Skewness is, in fact, a general problem in these data, and we attempted to correct for it by applying the transformation, $\log_e (x + 1)$, to all of the nondichotomous variables.

Timing of Measurement

The data for these measures of health and health-care utilization were obtained during the second, sixth, and tenth quarterly interviews for the head and spouse and during the second, eighth, and twelfth quarterly interviews for the children. An additional problem is presented by the children's twelfth quarterly interview which is not comparable to the earlier two because constraints of space limited the data collection to only the youngest and oldest children under 16. Consequently, we have excluded the twelfth quarter data from the analysis presented here. This timing presents a small problem for the analysis of the experiment's effects in that the second quarterly interview, in summarizing health data for the year prior to the interview, includes a period of six months prior to the start of the experiment during which the experiment obviously could have no effects. Thus, other things equal, the possibility of observing experimental effects is only half as likely at the second quarter.

Parameterization of Experimental Plans

Given that there are eight experimental plans combining the four guarantee levels and three tax rates, the problem arises as to how to best estimate the effects of these plans on our dependent variables.[7] Two different approaches were employed here. In the first, the eight plans were represented by a set of linear splines which progressively consider and exhaust the variation in the guarantee and tax rate through the introduction of terms that permit nonlinear and interactive effects.[8] However, as no nonlinear or interactive effects were found in our preliminary analyses, we shall report here only the effects of the three simplest spline variables that describe, respectively, the effects of membership in the experimental group (spline one), the effects of the guarantee level (spline two), and the effects of the tax rate (spline three). Spline one is simply a dummy variable scored one if the family is in the experimental group; zero otherwise. Spline two is defined as G-75, where G is the guarantee level for a particular family; spline three is defined as T-50, where T is a family's tax rate. Spline two tests for variation in the effect of the guarantee levels around a guarantee of 75 percent of the poverty line, while spline three tests for variation in the effect of the tax rates around the middle or 50 percent tax rate. Thus, when the three splines are present together in the same equation, spline one measures the effect of the plan with the 75 percent guarantee and 50 percent tax rate, and the other two splines estimate the effects of deviations from that plan in terms of guarantee and tax rate. The assumption underlying this parameterization is that the families have some genuine understanding of their guarantee level and tax rate and react to the experiment accordingly.

However, whether they actually have such an understanding can be questioned on both empirical and theoretical grounds. On the empirical side, Shore (1971) has found that knowledge of the experimental design was rudimentary in the sense that few families he surveyed could name their guarantee or tax rate; typically families knew little more than that their payments decreased as their earnings increased. On the theoretical side, these findings lend support to our suspicion that families may more readily respond to the actual payments they receive than to the payments they might potentially receive under their particular experimental plan. In suggesting this, we do not wish to imply any support for the notions of some culture of poverty theorists that the poor are

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

distinguished by a strong present time orientation or an inability to delay gratification, for the evidence on this question is highly moot (Miller, et al., 1968). Rather, we suggest that people generally prefer to act in terms of the concrete present—the payments—rather than the abstract future—the potential payments. Thus, our second parameterization is based on the actual payments received. To assess the possibility that the impact of the payments may occur only after a threshold amount is reached, the payments are expressed as a quadratic in the analysis.

We should note, however, that the most generous plans will, in general, make the largest payments.[9] Thus, although families may be explicitly making decisions regarding their health and utilization of care in terms of the size of the benefit check they receive, they will be implicitly reacting to a high guarantee and moderate to low tax rate—these being definitionally true of the most generous plans.

Control Variables

Because the design of the experiment did not randomly assign families to the various experimental plans and because the design was modified during the year-long period of screening and enrollment, several variables must be held constant in order to obtain unbiased estimates of the experimental effects. The most important of these is the income stratum of the family. The income stratum of a family is a gross estimate of its permanent income (the estimate was divided into three categories) conditioned by family size. Use of this model means that the various experimental plans differed in average family income at the beginning of the experiment, and consequently, the estimation of unbiased effects require that this variable be controlled for.

Ethnicity and experimental site must also be held constant because a decision was made to attempt to balance the sample ethnically during the period of screening and enrollment. As the four experimental sites were brought into the experiment sequentially over a year, this decision meant that certain ethnic groups were more heavily sampled in some cities than in others. For example, the Scranton sample is almost entirely white. Because ethnicity and site are, thus, necessarily confounded, the implication of this decision is that both must be held constant in any analysis. However, ethnicity, in any event, would have been held constant because of its importance in the literature on health and health care utilization (Mechanic, 1968).

In addition, the welfare status of the family must be controlled for because we are dealing with an experiment rather than a national program, and the experiment exists in a natural setting that includes a welfare program as an alternative to the experiment's benefits. The welfare problem exists in both states where there are experimental sites.[10] Moreover, in both states only those on welfare are eligible for Medicaid.

Several other factors, unrelated to design considerations, have been controlled in order to clarify the effects of the experimental plans. The age and education of the appropriate adult and family size, all measured at pre-enrollment, are held constant because of their aforementioned relationships with our dependent variables. The possession of health insurance, measured at the second or sixth quarter as appropriate, is also held constant for reasons given above. Average family earnings for the year (or six months in the case of the second quarterly interview) prior to the interview is also controlled for.

The estimates of effects for the spouse and the children each require one additional variable held constant that is inappropriate for the estimation of effects for the male head. With the exception of the equations for chronic conditions, a measure of whether the wife was pregnant in the year prior to the interview was included in the equations for the spouse, because for women of child-bearing age it is a potentially confounding factor in estimating the experiment's effects on the other dependent variables. In all the equations for the children a measure of the number of children below the age of six is included because of the greater use of medical care by children of that age (National Center for Health Statistics, 1968) and the aforementioned socioemotional

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

involvement of the parents with young children, which may lead to the discovery of chronic conditions. Also the mother's age and education are used for age and education in the children's equations.

Operational Models for Estimating Experimental Effects

In this section we shall describe the operational models, based on the sociological theory of a process leading from illness to the utilization of care, used to estimate the effects of the experiment on our dependent variables. We begin by noting that, although there are three points in time to be analyzed for the adults and two for the children, the first time point is to be distinguished from the others because there are no prior measures of health or utilization to be taken into account in the models. Consequently, in spite of the existence of three timepoints for the adults, there are, except for time differences, only two kinds of models: those that contain measures of health and utilization of care at an earlier point in time and those that do not. The latter, which occur at the second quarter, may be written as

$$
\begin{aligned}
C_2 &= f(E, X_2, e_f)\\
H_2 &= g(C_2, E, S_2, e_g)\\
W_2 &= h(H_2, C_2, E, X_2, e_h)\\
TD_2 &= i(W_2, H_2, C_2, E, X_2, e_i)\\
PD_2 &= j(W_2, H_2, C_2, E, X, e_j)\\
OD_2 &= k(W_2, H_2, C_2, E, X_2, e_k)
\end{aligned}
$$

where the subscript to the upper case letters refers to time and C = number of chronic illnesses; E = the experimental parameterization; X = the array of control variables; H = the number of days spent in a hospital (or at least one hospital stay for the children); W = the number of days lost from work (or the per capita number of days in bed for the children); TD = the total number of physician visits; PD = the total number of private physician visits; OD = the total number of other physician visits; and the e's are the appropriate disturbance terms.

The model asserts that, at the second quarter, a person's number of chronic illnesses is a function of only the control variables and his experimental status. This is consistent with the notion that a chronic condition is largely unaffected by the use of medical care. Second, the number of days in a hospital is taken as a function of the number of chronic illnesses, since many chronic conditions may require occasional hospitalization. Third, since hospitalization obviously implies that work days will be lost and since a chronic condition may be either occasionally or completely incapacitating, both these variables are assumed to affect work days lost. Finally, all three variables are stated as influences on physician visits; hospitalization implies physician visits; chronic illness and work days lost, if the underlying health condition is recognized as sufficiently serious, will also lead to physician visits.

Given the above model for the second quarter, we may write the following model for the later quarters:

$$
\begin{aligned}
C_{t+1} &= f'(C_t, E, X_{t+1}, e_{f'})\\
H_{t+1} &= g'(H_t, C_{t+1}, E, X_{t+1}, e_{g'})\\
W_{t+1} &= h'(W_t, H_{t+1}, C_{t+1}, E, X_{t+1}, e_h)\\
TD_{t+1} &= i'(TD_t, W_{t+1}, H_{t+1}, C_{t+1}, E, X_{t+1}, e_{i'})\\
PD_{t+1} &= j'(PD_t, W_{t+1}, H_{t+1}, C_{t+1}, E, X_{t+1}, e_{j'})\\
OD_{t+1} &= k'(OD_t, W_{t+1}, H_{t+1}, C_{t+1}, E, X_{t+1}, e_{k'})
\end{aligned}
$$

Since the prior values of the dependent variables are represented on the right, the equations estimate the effects of the experiment on changes in the dependent variables over time.

Because there are two different experimental parameterizations, six equations for each point in time, three points in time for each adult and two for the children, a total of 96 equations must be estimated in order to determine the effects of the experiment on the dependent variables. We cannot present, much less discuss, the coefficients for all 96 equations within the present limitations of space; consequently, we shall present here only the estimates for the experimental effects.[11]

The Sample

Our analysis focuses on the 732 husband-wife families which were in the experiment "continuously." A continuous, husband-wife family was defined as

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

family with husband and wife present that completed an interview at preenrollment, at the twelfth quarter, and at six of the remaining eleven quarters. In addition, the families had to be husband-wife and present at the second, sixth, and tenth quarters—the quarters at which health data were collected for the head and spouse—and they could have missed no more than two quarters in succession. Thus, families that split up for a period longer than two quarters or that attrited in terms of the above definition are not represented in the results reported here. The first exclusion is consistent with our desire to estimate effects for both the head and spouse, but the second is prompted simply by the fact that techniques for estimating the possible biases resulting from attrition have not yet been fully developed. The present sample represents 55 percent of all families originally enrolled.

RESULTS

We shall present the results for the husband, wife and children separately and, as previously noted, only the estimates for the two parameterizations of the experimental plans will be given. Recall that in the design parameterization we would expect the effects of the guarantee level to be positive and the effects of the tax rate to be negative, since an increasing guarantee and a decreasing tax rate both produce an increase in the size of the benefit paid to the family and hence increases their disposable income. However, expectations as to the signs of the coefficients in the payments parameterization are somewhat more complex. If there is no threshold effect—that is, the effect of the payments is linear—then the linear term should be positive and the quadratic term should be insignificantly different from zero. On the other hand, if there is a threshold effect, the linear term may have either a negative or a small positive sign while the quadratic term has a larger positive effect.

Experimental Effects on Family Members

Male head. Table 1 gives the results for male heads of households and they are easily summarized: regardless of the perspective from which they are viewed, there are no apparent experimental effects. First, if we search for coefficients with an appropriate sign and statistically significant at the five percent level, we find only four out of a possible 90 tests, a number which differs trivially from that which would be expected simply on the grounds of chance. Second, if we recall that the coefficients for guarantee and tax rate represent deviations from the 75–50 plan, we can see from the small size of the coefficients for the former two variables that the deviations are trivial. Third, inspection of the signs of the coefficients fails to reveal any pattern, let alone one consistent with our expectations.

Female head. Turning now to the data for the female heads of households displayed in Table 2, we see again that the experiment has had no effects. As in the case of the husbands, the number of statistically significant coefficients does not differ from chance, there is no pattern to the coefficients' signs, and the deviations from the 75–50 plan for either guarantee or tax rate are inconsequential.

Children. The data for the children who, it should be remembered, are the family members for whom effects are most likely, are given in Table 3. In general, there is once again almost no evidence that the experiment has had any effect. But a possible exceptional may exist for per capita total visits to the doctor at the second quarter. Here an apparently statistically significant negative tax-rate effect receives some support from the finding of significance for the payments parameterization. The finding suggests that a 25 percent increase in the rate means a reduction of almost one physician visit.

Aside from this "finding," the results are similar to those for the adults. The effects of the guarantee and tax rate differ trivially from the effects of the 75–50 plan; the coefficients show no particular pattern as to sign; and only ten of the 60 coefficients are statistically significant.

Since some researchers have argued that experimental effects might be slow to develop, it is worth noting that in none of these tables is there any more evidence of significant effects at the sixth, eighth, or tenth quarters than at the second.

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

TABLE 1. NET EXPERIMENTAL EFFECTS ON HEALTH VARIABLES AT THE SECOND, SIXTH, AND TENTH QUARTERS FOR MALE HEADS OF CONTINUOUS HUSBAND-WIFE FAMILIES

Health Variables	Effects of				
	75–50 Plan	Guarantee	Tax Rate	Payments	Payments Squared
A. Second Quarter					
Number of chronic illnesses	.002	.001	.001	−.049	.005
	(.08)	(1.62)	(.50)	(−1.08)	(1.18)
Number of hospital days	−.152	.003	.000	−.033	.001
	(−2.22)	(1.57)	(.06)	(−.31)	(.10)
Number of work days lost	.146	−.003	−.000	−.001	−.011
	(1.28)	(−.99)	(−.04)	(−.00)	(−.67)
Total physician visits	.052	−.001	.006	.115	−.013
	(.74)	(−.49)	(1.96)	(1.05)	(−1.28)
Private physician visits	.016	−.000	.003	.182	−.020
	(.22)	(−.18)	(.87)	(1.68)	(−2.00)
Other physician visits	.029	−.001	.006	−.031	.002
	(.56)	(−1.04)	(2.70)	(−.38)	(.24)
B. Sixth Quarter					
Number of chronic illnesses	.015	−.000	.001	−.007	.001
	(.59)	(−.69)	(.92)	(−.27)	(.35)
Number of hospital days	−.940	.009	−.030	.278	−.037
	(−1.36)	(.54)	(−.98)	(.39)	(−.84)
Number of work days lost	.037	.003	−.002	.054	−.006
	(.31)	(.93)	(−.29)	(.44)	(−.83)
Total physician visits	−.291	.003	−.019	.087	−.034
	(−.43)	(.19)	(−.64)	(.12)	(−.80)
Private physician visits	.625	−.002	−.100	.143	−.037
	(1.01)	(−.15)	(−.37)	(.22)	(−.95)
Other physician visits	−.826	.005	−.011	−.177	.012
	(−2.58)	(.69)	(−.77)	(−.53)	(.58)
C. Tenth Quarter					
Number of chronic illnesses	.002	.001	.001	−.049	.005
	(−.40)	(.80)	(1.40)	(−.01)	(.38)
Number of hospital days	−.609	−.018	.049	.600	−.030
	(−.51)	(−.61)	(.93)	(.51)	(−.43)
Number of work days lost	.177	.001	.002	−.386	.020
	(1.45)	(.43)	(.33)	(−3.23)	(2.83)
Total physician visits	.969	−.006	.032	−.301	.020
	(1.45)	(−.39)	(1.08)	(−.45)	(.49)
Private physician visits	.542	−.010	.024	−.364	.021
	(.92)	(−.68)	(.93)	(−.63)	(.60)
Other physician visits	.200	.005	.007	.032	.002
	(.78)	(.73)	(.63)	(.13)	(.12)

Note: These are unstandardized regression coefficients with t-values in parentheses. See text for variables held constant in making estimates. The effects of the guarantee and tax rate are relative to the 75–50 plan. Payments are expressed in thousands of dollars per quarter.

CONCLUSIONS

In summary, we may say that the experiment had no effects on either our measures of health or on our measures of the utilization of health care. That this is the case should not be surprising, given that review of the literature found that the effects of income on these variables were generally small or nonexistent. The ex-

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

TABLE 2. NET EXPERIMENTAL EFFECTS ON HEALTH VARIABLES AT THE SECOND, SIXTH AND TENTH QUARTERS FOR FEMALE HEADS OF CONTINUOUS HUSBAND-WIFE FAMILIES

Health Variables	Effects of				
	75–50 Plan	Guarantee	Tax Rate	Payments	Payments Squared
A. Second Quarter					
Number of chronic illnesses	−.015	−.000	.002	.065	−.004
	(−.50)	(−.03)	(1.75)	(1.38)	(−1.04)
Number of hospital days	−.210	.003	.001	−.019	.005
	(−2.54)	(1.47)	(.36)	(−.15)	(.45)
Number of work days lost *	.033	.003	.007	−.008	−.005
	(.30)	(1.25)	(1.44)	(−.05)	(−.29)
Total physician visits	.018	.002	−.001	.073	−.005
	(.22)	(.76)	(−.28)	(.59)	(−.44)
Private physician visits	−.029	.001	−.003	.218	−.018
	(−.33)	(.39)	(−.80)	(1.63)	(−1.45)
Other physician visits	.071	.000	−.001	−.078	.006
	(1.22)	(.25)	(.55)	(−.88)	(.78)
B. Sixth Quarter					
Number of chronic illnesses	.051	−.000	−.000	.047	−.003
	(1.89)	(−.42)	(−.03)	(1.67)	(−1.84)
Number of hospital days	.060	.007	−.012	.716	−.047
	(.12)	(.57)	(−.56)	(1.38)	(−1.49)
Number of work days lost	.109	−.001	−.005	−.018	.004
	(1.10)	(−.53)	(−1.17)	(−.18)	(.65)
Total physician visits	−.101	−.003	.073	−.437	−.032
	(−.15)	(−.16)	(2.53)	(−.63)	(−.76)
Private physician visits	−.289	−.011	.054	−.458	.027
	(−.46)	(−.72)	(1.94)	(−.69)	(.69)
Other physician visits	.253	.009	.020	−.088	.010
	(.89)	(1.28)	(1.60)	(−.29)	(.57)
C. Tenth Quarter					
Number of chronic illnesses	−.036	.001	.001	−.019	.001
	(−1.34)	(1.03)	(.95)	(−.70)	(.70)
Number of hospital days	−.216	.013	−.017	1.36	−.061
	(−.29)	(.74)	(−.50)	(1.83)	(−1.38)
Number of work days lost	−.025	.003	−.001	−.195	.009
	(−.23)	(1.13)	(−.23)	(−1.83)	(1.41)
Total physician visits	−.757	−.008	−.035	−1.54	.095
	(−1.05)	(−.42)	(−1.10)	(−2.16)	(2.23)
Private physician visits	−.333	.003	−.033	−.901	.067
	(.49)	(.20)	(−1.12)	(−1.34)	(1.66)
Other physician visits	−.354	−.009	.001	−.669	.031
	(−1.28)	(−1.30)	(.09)	(−2.46)	(1.91)

See *Note* for Table 1.
* Includes housework.

perimental payments are, after all, only addition to the income of our families, and we should not be surprised if they use this added income in ways similar to those in which they use their other income.

It can, of course, be argued that our failure to find any experimental effects is due to the experiment's limited duration, or to our sample's lack of access to care, or to meagerness of the experimental payments. However, we do not find any of these arguments, individually or taken

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

TABLE 3. NET EXPERIMENTAL EFFECTS ON HEALTH VARIABLES AT THE SECOND AND EIGHTH QUARTERS FOR CHILDREN OF CONTINUOUS HUSBAND-WIFE FAMILIES

Health Variables	Effects of				
	75–50 Plan	Guarantee	Tax Rate	Payments	Payments Squared
			A. Second Quarter		
Per capita chronic illnesses	.038 (1.24)	−.001 (−1.49)	.000 (.13)	.054 (1.14)	−.005 (−1.11)
Hospital stay	−.091 (−2.70)	.002 (2.33)	−.001 (−.55)	.082 (1.57)	−.007 (−1.55)
Per capita bed days	.028 (.30)	−.000 (−.13)	.001 (.26)	.042 (.29)	−.004 (−.31)
Per capita total physician visits	−.054 (−1.21)	.002 (1.45)	−.004 (−2.23)	−.141 (−2.08)	.016 (2.54)
Per capita private physician visits	−.029 (−.62)	.002 (1.48)	−.004 (−2.23)	−.089 (−1.26)	.010 (1.58)
Per capita other physician visits	−.053 (−1.77)	.000 (.18)	.001 (.53)	−.061 (−1.32)	.007 (1.72)
			B. Eighth Quarter		
Per capita chronic illnesses	.014 (.65)	.000 (.59)	.001 (.74)	−.012 (−.49)	.000 (.02)
Hospital stay	.050 (1.51)	.001 (.92)	−.001 (−.64)	.099 (1.09)	−.005 (−.86)
Per capita bed days	.097 (1.10)	−.002 (−.90)	−.005 (−1.25)	−.003 (−.09)	.000 (.19)
Per capita total physician visits	.023 (.46)	−.001 (−.78)	.002 (.78)	−.049 (−.95)	.002 (.76)
Per capita private physician visits	.044 (.88)	−.001 (−.76)	.000 (.16)	−.032 (−.63)	.001 (.38)
Per capita other physician visits	.009 (.22)	−.000 (−.25)	.004 (2.13)	−.058 (−1.43)	.004 (1.46)

See *Note* for Table 1.

together, at all compelling. While the experiment was limited to only three years, the fact that people generally put an extremely high priority on their health leads us to suspect that the effects—if they were to be observed at all—should have occurred quite rapidly. Moreover, the utter lack of any sign of an increase in illness or utilization over time makes it unlikely that a longer experiment would produce different results. As far as the question of access to care is concerned, we have no evidence that our sample is seriously disadvantaged in this respect. The vast majority of the families report that they have a regular physician within a reasonable distance from their homes. Finally, the argument that the experimental payments are too small to have any effect can be dismissed with the simple observation that the average payment per family per year was about $1,000; since average family earnings for the whole sample were approximately $5,500, this amounts to 18 percent of their earnings—a nontrivial amount.

In conclusion, we would like to point out once again that despite the lack of experimental effects, these findings are consistent with previous research on income and health which indicates that health and medical care are more a function of life style (preferences) than income—at least beyond the level of mere subsistence (Lefcowitz, 1973a; 1973b). Since the experiment was not designed to change life styles, the outcomes reported here were to be expected. Yet it is worth

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

noting that the results were not expected in terms of economic theory and, while sociological research provided some basis for anticipating them, it nonetheless also contained no model that permitted their prediction.

NOTES

1. Introduced by Milton Friedman, the negative income tax has been held to be superior to existing welfare schemes because its tax on earnings is less than 100 percent—thus ostensibly providing an incentive to work absent in existing welfare; its benefits would be paid as a simple function of income rather than in terms of eligibility, in terms of a set of categorical rules, and payments would not be subject to the discretion of caseworkers.
2. Perhaps it should be said that improvement in the ill depends upon both the efficacy of care and the belief of the ill in its efficacy. The efficacy of care on the level of health is increasingly being questioned; see, for example, Benham and Benham, 1975).
3. This statement should be qualified by the fact that Richardson found that persons on Medicare used physicians significantly *less* than those on voluntary insurance; however, this exception is not really germane to the present discussion as our sample is almost entirely ineligible for Medicare. It should also be noted that there was an insurance effect when the condition was not serious.
4. For reasons of space, this discussion is necessarily brief. For greater details, see Part C of the *Final Report of the Graduated Work Incentive Experiment* in New Jersey and Pennsylvania available from the Institute for Research on Poverty, University of Wisconsin, Madison, Wisconsin 53706. The *Report* will also be published by Academic Press during the coming year.
5. Guarantee levels were annually adjusted for changes in the cost of living.
6. This and the following correlation were computed from logarithmitically transformed variables, as described below.
7. Recall that the eight plans represent the subset of the 12 possible combinations of guarantee and tax rate of primary policy interest.
8. We shall not attempt to introduce here the theory of splines to those unfamiliar with the technique as we are making only modest use of it. Readers desiring further background should consult Poirier (1973).
9. The potential generosity of a plan can be expressed in terms of its breakeven point (the point at which payments cease), which can be found by dividing its guarantee level by its tax rate.
10. For an analysis of the impact of welfare, see Garfinkel (1974).
11. A full set of tables is available upon request from the authors.

REFERENCES

Aday, Lu Ann and Robert Eichorn
1972 The Utilization of Health Services: Indices and Correlates. Washington, D.C.: National Center for Health Services Research and Development, Department of Health, Education, and Welfare.

Andersen, Ronald
1968 A Behavioral Model of Families' Use of Health Services. Research Series No. 25. Chicago: Center for Health Administration Studies, University of Chicago.

Andersen, Ronald and Lee Benham
1970 "Factors affecting the relationship between family income and medical care consumption." Pp. 73-95 in Herbert E. Klarman (ed.), Empirical Studies in Health Economics. Baltimore: The Johns Hopkins Press.

Andersen, Ronald, Joanna Kravits, and Odin W. Anderson (eds.)
1975 Equity in Health Services. Cambridge: Ballinger.

Burham, Lee and Alexandra Benham
1975 "The impact of incremental medical services on health status, 1963–1970," Pp. 217–28 in R. Andersen, J. Kravits and O. W. Anderson, (eds.), Equity in Health Services. Cambridge: Ballinger.

Cole, S. and R. LeJeune
1972 "Illness and the legitimation of failure." American Sociological Review 37:347–56.

Conlisk, John and Harold W. Watts
1969 "A model of optimizing experimental designs for estimating response surfaces." Proceedings of the Social Statistics Section. Washington, D.C.: American Statistical Association.

Elesh, D. and Schollaert, P.T.
1972 "Race and urban medicine: Factors affecting the distribution on physicians in Chicago." Journal of Health and Social Behavior 13:236–50.

Feldman, Jacob J.
1966 The Dissemination of Health Information. Chicago: Aldine.

Friedman, Milton
1957 A Theory of the Consumption Function. National Bureau of Economic Research. Princeton, N.J.: Princeton University Press.

Garfinkel, Irwin
1974 "The effects of welfare programs on experimental responses." Journal of Human Resources 18:504–29.

Harkey, J., D. L. Miles and W. A. Rushing
1976 "The relation between social class and functional status: A new look at the drift hypothesis." Journal of Health and Social Behavior 17:194–204.

Koos, Earl
1954 The Health of Regionville. New York: Columbia University Press.

Lefcowitz, M.J.
1973a "Poverty and health: A re-examination." Inquiry X:3–13.
1973b "Poverty and Health: Children's Medical

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions

Care." Discussion Paper No. 159–73. Madison, Wisconsin: Institute for Research on Poverty, University of Wisconsin.

Luft, Harold S.
1972 "Poverty and health: An empirical investigation of the economic interactions." PhD. dissertation. Boston: Harvard Center for Community Health.

Mechanic, David
1968 Medical Sociology: A Selective View. New York: Free Press.

Miller, S.M., Frank Riessman and Arthur A. Seagull
1968 "Poverty and self-indulgence: A critique of the non-deferred gratification pattern." Pp. 416–32 in Louis A. Ferman, Joyce L. Kornbluh and Alan Haber (eds.), Poverty in America. Ann Arbor: University of Michigan Press.

National Center for Health Statistics
1968 "Volume of Physician Visits by Place of Visit and Type of Service." Vital and Health Statistics, Series 10, No. 49. Washington, D.C.: U.S. Department of Health, Education and Welfare.
1972a "Age Patterns in Medical Care, Illness, and Disability." Washington, D.C.: Vital and Health Statistics, Series 10, No. 70. U.S. Department of Health, Education, and Welfare.
1972b "Hospital and Surgical Insurance Coverage." Vital and Health Statistics, Series 10, No. 66. Washington, D.C.: U.S. Department of Health, Education, and Welfare.

Poirier, Dale J.
1973 "Multiple Regression Using Bilinear Splines." Social Systems Research Institute Workshop Paper No. 7230. University of Wisconsin Social Systems Research Institute, Madison, Wis.: University of Wisconsin.

Richardson, William C.
1971 Ambulatory Use of Physicians' Services in Response to Illness Episodes in a Low Income Neighborhood. Chicago: Center for Health Administration Studies, University of Chicago.

Samora, J., L. Saunders and R.F. Larson
1962 "Medical vocabulary knowledge among hospital patients." Journal of Health and Social Behavior 2:83–92.

Shore, Arnold
1971 "Institutional Assistance: A study of the negative income tax experiment." Ph.D. dissertation. Princeton, N.J. Princeton University.

Shortell, Stephen M.
1975 "The effects of patterns of medical care on utilization and continuity of services," Pp. 191–216 in R. Andersen, J. Kravits and O.W. Anderson (eds.), Equity in Health Services. Cambridge: Ballinger.

RESEARCH GRANTS AND DRUG SELF-HELP PROGRAMS: WHAT PRICE KNOWLEDGE? *

LINCOLN J. FRY
Institute for Urban Affairs and Research, Howard University

Journal of Health and Social Behavior 1977, Vol. 18 (December):405–417

Based upon 14 months of participant observation in a patient-run drug rehabilitation program located in a mental hospital, this paper is concerned with the impact of certain kinds of funding arrangements upon self-help programs. The findings reported support Cloward and Piven's (1974) contention that large research grants reflect the interests of professionals, not those of the groups that the projects are supposed to serve. Grant-funded activities eventually coopted a self-help program's leadership, and research infringed upon program activity. Funding priorities alienated the institution's treatment staff and contributed to competition between the self-help program and the other drug units located in the institution's treatment complex. The implication of this case study is that self-help programs are not likely to maintain their organizational integrity when implemented by the use of large research grants.

Cloward and Piven (1974) maintain that large scale funding of research connected with social service ventures has resulted in the development of programs which further the careers of various classes of professionals but which provide only meager benefits to the disadvantaged groups the programs purport to serve. From Cloward and Piven's perspective, large scale funding creates bureaucratic structures which become the domain of professionals and their special knowledge, a process the authors define as "the con-

* The comments and suggestions provided by Judith Lorber and Mary E. W. Goss on an earlier version of this paper are gratefully acknowledged.

This content downloaded from 130.63.180.147 on Sat, 12 Dec 2015 00:38:33 UTC
All use subject to JSTOR Terms and Conditions